Unit 3 · Outcome 2 · Sustainable Development

Who decides how much water the wetland gets?

KK06 VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.

A decision sits at the centre of a web

02 The four factors A decision sits at the centre of a web Every environmental decision at Boorla is pushed and pulled by the same four factors. Drag to orbit. Click a factor to light up its links to the others. Three.js · interactive Decision web · green links = interconnections · red links = tensions All links Interconnections Tensions Click a node to isolate its links. VCAA names exactly four factors. Memorise them as S·R·D·T — and notice that two are about people and two are about evidence . Stakeholder values, knowledge & priorities What different g

Interconnections and tensions — learn the difference cold

03 The spine of the dot point Interconnections and tensions — learn the difference cold The two words in the dot point are the whole game. An answer that only lists the four factors will sit on low marks. The marks live in showing how they relate . Interconnection Where two factors reinforce or depend on each other — pulling the same way. Example: new sensor technology generates the scientific data that regulatory targets are built on. Move one, the other moves with it. Tension Where two factors pull against each other — satisfying one strains the other.

Four groups, one river

04 Factor S · the people Four groups, one river Drag the slider to send more of the year's water down to the wetland as an environmental flow . Watch every group's satisfaction respond — you cannot peak all four at once. That impossibility is the tension. Three.js · interactive Boorla catchment · river red gums green up as environmental flow rises Environmental flow to wetland 35% of flow The four stakeholder groups Irrigators & horticulture Priority: secure, affordable water for dairy and crops near Connawarra. Value system leans anthropocentric — the r

Regulatory frameworks set the boundaries

05 Factor R · the rules Regulatory frameworks set the boundaries Managers at Boorla cannot simply do what the science suggests or what one group wants. They operate inside a stack of binding rules — international, national, state and local. These frameworks inform and constrain the management strategy. Ramsar Convention (international) — Boorla is a listed wetland of international importance; Australia is obliged to maintain its "ecological character". This elevates the wetland's claim on water. Water Act 2007 (Cth) & the Murray–Darling Basin Plan — set

Historical and current scientific data

06 Factor D · the evidence Historical and current scientific data VCAA is deliberate about the words "historical and current". Responsible decisions compare what the system used to do with what it is doing now . Historical data Long-term flood records, old aerial photographs, fish and bird counts going back decades, and Traditional knowledge of the wetland's natural flood rhythm. It sets the baseline — what "healthy" looked like. Current data Real-time flow gauges, recent vegetation-condition surveys, water-quality probes and satellite greenness. It show

New technologies — the double-edged factor

07 Factor T · the tools New technologies — the double-edged factor Technology can dissolve a tension or manufacture one. That double edge is what makes it examinable. Eases tensions Drip irrigation & soil-moisture sensors let farms grow the same crop with less water — freeing volume for the wetland. Fishways let Murray cod pass the dam. Satellite greenness (NDVI) cheaply monitors recovery. Creates tensions New tech costs money — who pays? (the user-pays principle). Efficient irrigation can encourage more planting (the "rebound effect"), erasing the water

A worked decision: the spring environmental flow

08 Putting it together A worked decision: the spring environmental flow The CMA proposes releasing a managed spring flood from Yarrin Dam to trigger river red gum growth and waterbird breeding. Watch all four factors switch on at once — some reinforcing, some resisting. Scientific data (D) shows canopy condition has crossed a stress threshold and birds haven't bred in three years — a trigger for action. Regulatory frameworks (R) make the flow possible : the Basin Plan reserves the environmental water, and Ramsar obliges Australia to protect ecological ch

Command-word decoder

◈ **COMMAND WORDS**

09 Decode the question Command-word decoder The same content earns different marks depending on the command word. Tap a card to see what the examiner actually wants for this dot point.

Six named exam traps

⚠ **EXAM TRAPS**

10 Where marks leak away Six named exam traps T1 Listing factors without linking them Naming S, R, D, T earns recall marks only. Fix: for every factor, state one interconnection or tension with another. T2 Treating "interconnection" and "tension" as the same word They're opposites. Fix: interconnection = reinforcing; tension = conflicting. Use both words explicitly. T3 Generic stakeholders "The community" is vague. Fix: name the Boorla groups and their specific priority and value system. T4 Forgetting technology can be negative Students cast tech as a pu

P·E·L writing trainer

✎ **P·E·L WRITING**

11 Write like a state-ranker P·E·L writing trainer Structure every extended-response paragraph as Point → Evidence → Link . The prompt below is a typical 4-mark "analyse" task. Prompt · Analyse one tension between two factors in the Boorla decision (4 marks) Build the answer one move at a time, then reveal the model. P · POINT State the tension in one sentence, naming the two factors. E · EVIDENCE Anchor it in the case study with a specific, named example for each factor. L · LINK Explain the consequence for the decision and tie it to a sustainability pr

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

VCE-style practice questions

★ **PRACTICE & SELF-MARK**

12 Practice & self-mark · 20 marks VCE-style practice questions Try each on paper first. Then reveal a weak answer, an annotated strong answer and the marking guide, and self-mark. Your total feeds the dock. Q1 Define what is meant by a stakeholder in environmental decision-making, and identify two stakeholder groups at the Boorla Wetlands with differing priorities. 2 marks define / identify 👁 Reveal answers ✗ Weak · 0–1/2 "A stakeholder is someone involved. Two groups are the farmers and the environment." Definition is too thin; "the environment" is not

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.

P · POINT	E · EVIDENCE	L · LINK
answer the verb	example / mechanism	tie back to question

The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

A stakeholder is someone involved. Two groups are the farmers and the environment." Definition is too thin; "the environment" is not a stakeholder group, and no priorities are given.

✓ STRONG / MODEL ANSWER

A stakeholder is any individual or group with an interest in, or affected by, an environmental decision . Two groups with differing priorities are Connawarra's irrigators, who prioritise secure water for farming , and the Boorla Traditional Owners, who prioritise cultural flows that flood the wetland ." Crisp definition + two named groups + a contrasting priority each.

✗ WEAK ANSWER

There are laws that protect the wetland so they have to give it water." Single vague point; no named framework, no mechanism. Caps at 1.

✓ STRONG / MODEL ANSWER

Regulatory frameworks both enable and require the flow. The Murray–Darling Basin Plan reserves a legal volume of environmental water , delivered by the Commonwealth Environmental Water Holder, which makes the release possible. The Ramsar Convention obliges Australia to maintain the wetland's ecological character , creating a duty to act. Meanwhile existing irrigation entitlements constrain how much can be reallocated, so the frameworks also set the upper limit." Three distinct, named frameworks each with a clear influence — enabling, requiring, constraining.

✗ WEAK ANSWER

An interconnection is when things connect and a tension is when they disagree. The farmers and the wetland have a tension." Defines loosely, gives only one example, no interconnection example, no clear contrast.

✓ STRONG / MODEL ANSWER

An interconnection is where two factors reinforce or depend on one another : for example, new sensor technology generates the scientific data the CMA uses to set watering targets — the two factors strengthen each other. A tension is where two factors pull in opposite directions : for example, irrigators' priority for maximum allocation conflicts with the Basin Plan's environmental-flow requirement , so a gain for one is a loss for the other. The key difference is that an interconnection aligns factors while a tension opposes them." Both terms defined, both illustrated with a distinct named example, and an explicit point of distinction.

✗ **WEAK ANSWER**

The bird data is bad so they should fix the swamp. The farmers don't want to pay more. There are rules about water quality." Spots three factors but doesn't name them, doesn't interpret the data conflict, and never links or weighs them.

✓ **STRONG / MODEL ANSWER**

Scientific data pulls two ways: twenty years of survey data (historical) shows a breeding collapse , justifying action, yet the recent wet winter (current) could be misinterpreted as recovery — a reason to apply the precautionary principle. Stakeholder priorities create a tension : oyster farmers prioritise low costs, raising a user-pays/equity issue against the swamp's ecological need. New technology (recycled-water piping) makes the intervention possible but introduces a cost. The regulatory framework (state water-quality standard) constrains the scheme, interconnecting with the data because approval depends on monitoring. A responsible decision weighs the long-term ecological evidence ab

✗ **WEAK ANSWER**

All four factors matter. Stakeholders disagree, there are laws, data is collected and technology helps. The decision should be sustainable for the future." Mentions all four but with no examples, no interconnection/tension analysis and only a token nod to principles. Stalls in the middle band.

✓ **STRONG / MODEL ANSWER**

The four factors are deeply interwoven. Scientific data (D) — river red gum canopy-condition surveys and three failed breeding seasons — triggers the proposal, and is interconnected with regulatory frameworks (R) : the Basin Plan and Ramsar both rely on that evidence to justify reserving environmental water. Stakeholder values (S) create tension : the Boorla Traditional Owners and the CMA (ecocentric) support the flood, while some irrigators (anthropocentric) fear reduced allocation — a tension within the single factor S. New technology (T) — soil-moisture sensors and drip irrigation — eases that S–R tension by cutting on-farm demand, though the cost raises a user-pays issue. Weighing these,